Education using Virtual Reality for students with learning disabilities

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***Abstract*—Visual Education Resources (VISER) is a web based Virtual Reality (VR) platform which is created to target people suffering from dyslexia and to provide them with an educative, interactive and easily accessible learning platform. This paper mainly discusses the implementation of the proposed system and an insight of how Natural Language Processing (NLP) has been used for Automatic Text Visualization (ATV) along with Web Virtual Reality (WebVR). The paper further explains how 3D avatars can be incorporated in WebVR along with their animations.**

Keywords— ***VISER, Dyslexia, VR, WebVR, NLP, ATV***

I. INTRODUCTION

Dyslexia is a specific learning disability which causes problems such as learning and cognition. It is a disorder that usually affects a persons' ability to read, spell and comprehend certain things. If dyslexia is diagnosed in the early stages, then it can be cured very effectively by giving proper guidance and support to the dyslexic person. Dyslexia is currently one of the growing issues among the young generation. According to a survey it is observed that around 5-10 percent of Americans suffer from dyslexia. This disability can be cured through proper guidance, support and psychological testing which is generally a special learning program for teaching dyslexic people.

The traditional techniques to treat dyslexia require personal attention to be given to each individual student. Special classes need to be conducted where the students are taught using the classic chalk and board techniques. These techniques of teaching are used in almost all dyslexia treating organizations, but these techniques have some limitations.

In order to overcome the condition of dyslexia and improve traditional teaching techniques we are proposing a system "Visual Education Resources (VISER)" which supports the above-mentioned title "Education for Dyslexic Students using Virtual Reality". This system is a Web Virtual Reality (WebVR) based education platform for dyslexic students. In [3] the researchers have used the technique of virtual reality in order to teach students using virtual representation. This was our main motivation towards developing this system. It is observed that dyslexia can be cured faster if the students are taught by using some new and innovative teaching techniques. We are developing a system where we will teach the students in a virtual environment which will attract the students and gain their attention towards learning. The system will be an immersive learning platform for students where they will be present in a virtual world surrounded by various modules through which they can start learning.

The following section describes the overall architecture of the proposed system. The third section gives a detailed idea about the various algorithms used in the system. Next section which is section four gives an idea about the system along with the technologies used for development. After that next section gives brief idea about performance evaluation using syntax rules. In the further section, results of the system are portrayed. Finally, in the last section the conclusion and future work related to the system.

II. PROPOSED SYSTEM

In the following sections, we have described the overall system architecture along with algorithms used for text visualization in virtual world using natural language processing. Apart from this, we have also shown the implementation of system along with examples and results.

A. SYSTEM ARCHITECTURE

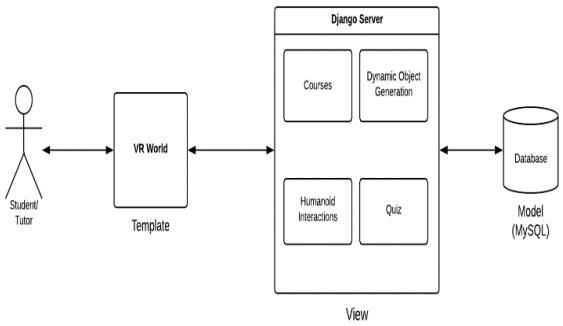


Fig. 1. System Architecture for VISER

This is an overall architecture of our system which is the basic Model Template View (MTV) type architecture. The user is the teacher or the student interacting with the system. The template contains the VR world developed for the users in order to interact with the system. The view contains the

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backend related to the entire system which is mainly used for the processing purpose. The model contains the database which is repository for storing various assets that will be used by our system. In the following section the different modules in VISER are emphasized.

In our system we have developed four modules viz. courses, Dynamic Object Generation (DOG), humanoid interaction and quiz. A brief overview of the modules is explained further. After reading [19] we found ideas on how to teach students suffering from learning disabilities. Using these ideas we have developed the course module. In the course module various courses are provided to the students through which they can clear their core concepts. The various courses provided in our system are shapes, colors and quantity. The courses are present in the virtual environment, so it makes the learning process for the students much more fun. For example, in the quantity course there are number scattered in the virtual world all around the screen and the student can select a number from the numbers present. After selecting a number, a 3D object will be displayed on the screen that many specified number of times. For example, if we select the number 2 then two apples will be rendered on the screen. The students can roam anywhere around in the virtual world in order to get a clear picture of what they are seeing. This increases the learning ability of the students as a person learns more quickly through visual memory.

The quiz module is used to test the students in order to gain an insight of the knowledge the student possess. In our project we have developed a static quiz which is a mixture of questions which test the students in various fields. The quiz is also developed in the virtual environment and contains various objects and questions related to the objects. The student can have a 360 view of the objects and then answer the question which makes the test much more interactive and fun as compared to the traditional pen and paper tests. At the end of the test a scorecard is generated which gives the analysis as to how the student has performed in the test.

In [10], a system which generates a 3D scene based on a text provided to the system has developed. We are using this idea to develop a module in our system called the DOG. DOG is the module which stresses on the term's cognition and spatial relations. This feature was developed because complex sentences such as 'There is a small brown ball on the right side and a huge elephant is far on the left side' is difficult for the dyslexic student to break down and comprehend. This module takes this input and then applies NLP on it and then returns a rendered 3D scene in the VR world. The scene will generate the objects along with their respective attributes and maintains the spatial relations among these objects. In short, this module will enhance the cognitive and imaginative power of the dyslexic student. DOG module also handles manipulation of scale, position and color of these 3D objects.

Humanoid action is the feature provided in VISER wherein an actual virtual avatar pops up in the VR world and carries out the mentioned actions. This module mainly focuses on the verbal actions that humans see in everyday life like jumping, dancing, and running, etc. If the user types in the query like 'Show me how to dance', the text is processed, and the corresponding action of the current selected avatar is played. An added subsidiary of humanoid actions is the ability for the user to select and use his own avatar, also known as user preferences. This module also allows the user to explicitly call another avatar temporarily without selecting the avatar, for e.g. 'Maya is dancing'. Here 'Maya' is explicitly called to be the temporary avatar whereas in a query like 'Show me how to punch', the default selected avatar for that specific user is called for use.

In [21], the author has carried out an extensive literature survey in the field of VR. This survey conducts deep research on domain like virtual reality, text visualization and education through VR, which has formed the base for the developed system.

B. ALGORITHM

This section provides the step by step detailed processing of the two modules namely – DOG and Humanoid actions

(a) DOG

The algorithm for DOG module is as follows:

The user first enters the query for the scene he/she wants to visualize. The first step in the algorithm will be pre-processing the query. It contains removal of stop words, tokenization of query. After this, we apply Part-Of-Speech (POS) tagging to the query. Following this step, total number of nouns i.e. objects to be rendered is counted.

We will be representing the object (3D object to be rendered) in the form of tuple like this after processing the query.

1. {[Noun, size, color, X, Y]}

The above representation is valid only if query contains single object to be rendered.

For more than one object the tuple representation is as follow.

{'noun_1': [size, color, X, Y], 'noun_2': [size, color, X, Y]}

Noun – The noun of 3D object to be rendered.

Size – It includes extra small, small, medium, large and extra-large etc.

Color – It contains color of an object. (red, green, blue, etc.)

X - Spatial direction (left, right, above, below, up, down)

Y - Spatial distance (far, near, away)

Once tuple is created, we have to fetch the 3D files i.e. .obj and .mtl files from database along with its co-ordinates. So, the files and corresponding co-ordinates are fetched, converted into list and passed to the template where it renders the scene.

Ex1. Show me a green apple on the left side of the brown table.

Here green is associated with apple, brown is associated with table. Left is associated with apple. So, tuple will be like this:

{'apple': [None, green, left, None], 'table': [None, brown, None, None]}

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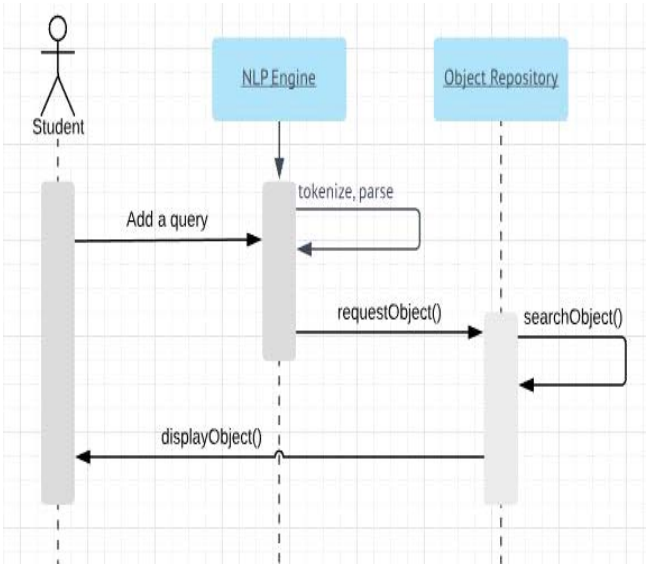


Fig. 2. Process flow for DOG

Ex2. Show me an elephant on the left side which is away.

Here only spatial direction & distance associated to elephant is being mentioned. Therefore, the tuple will look like: {[elephant, None, None, left, far]}

Ex3. Show me an apple and a table.

If the spatial distance and direction both are absent in query, then the objects will be rendered from left to right sequence as mentioned in query. Here apple will be on the left side of table. The tuple formed will be: {'apple': [None, None, None, None],

'table ': [None, None, None, None]}.

Ex.4 Show me a laptop which is up and far from me.

Here the tuple formed will look like: {[laptop, None, None, up, far]}

Ex.5 Show me an apple, car and table.

Here the tuple formed will look like: {'apple': [None, None, None, None], 'car: [None, None, None, None], 'table': [None, None, None, None]}

Ex.6 The apple is kept on the table. On the other hand, a big car is on the left side and away from me.

Here the tuple formed will look like: {'apple': [None, None, On, None], 'table': [None, None, None, None], 'car': [big, none, left, away]}

The above example no.6 is special case. In our algorithm for some specific objects, we have stored the relative co-ordinates between the two mentioned objects. Whenever these special conditions are encountered, we retrieve the co-ordinates from a different table in our database.

Unless and until size is explicitly mentioned it will be considered as medium size i.e. default size for every object. And now as size and few parameters are not mentioned it will be None i.e. default stored value will be considered for them.

As in some cases, spatial relations are explicitly mentioned so, we must alter its co-ordinates in reference of the mentioned spatial relationship.

In the tuple, if 'up' is associated with object 'X' then we have manipulated the y-axis co-ordinate of object 'X' while rendering the object. Similarly, if 'far'/'near' is mentioned then we have manipulated its z-axis co-ordinate while rendering the object and in the same context for 'left'/'right' we have manipulated x-axis co-ordinates.

We have divided 3D objects in 5 categories based on its real-life sizes. The five categories are, extra small, small, medium, large, extra-large. For a particular object, if the size attribute is not None then the size of that specified object is manipulated based on some tested and generalized values based on the category of an object.

Ex5. Show me a big red balloon.

The tuple formed will be: {[balloon, big, red, None, None]}

The balloon has mentioned size which is big. The balloon belongs to category small. So, while rendering balloon the scale attribute will be manipulated based on some tested and stored value which are generalized for category small.

(b) HUMANOID ACTIONS

The algorithm for Humanoid action module is as follows:

The user will enter the text query for the actions he/she wants to visualize. The first step is to pre-process the user entered query. It involves removing of stop words from the query. Then we have applied stemming technique which converts the word into its base form. Ex. 'running' gets stemmed to 'run'.

We have designed a static unigram POS tagger to identify the names of an avatar and associated actions. Thus, we applied our unigram tagger to the query and obtain the tagged_query. In the user preference section, we asked user to select an avatar of his/her choice. Then the user preferred avatar will be used for the further processing.

Ex1. Show me how to dance.

Suppose user has selected 'Akshay' as his preferred avatar in user preference section. Then the output will be Akshay is dancing.

Ex2. Mohan is doing exercise.

Suppose here user's preferred avatar is 'Maya'. But as user has mentioned character's name in the query, so the output will be avatar of 'Mohan' doing exercise.

Ex3. Akshay is kicking and Mohan is dancing. On the other hand, I am punching.

Suppose user has selected 'Akshay' as his preferred avatar in user preference section. Then the output will be on the left most side Akshay is kicking, In the middle Mohan is doing dancing action, and on the right most side Akshay is doing punching action.

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We have created folders by the name of avatars on the server. In those folders we have fbx animation files associated with respected avatars. So, when query is fired 'Mohan is doing exercise' then \mohan\exercise.fbx gets called and executed. In database we have stored avatar name with scale and position associated with respective avatars. So, whenever an avatar gets rendered, first its scale and position is fetched from database. Then with the help its scale, position and .fbx file an animation is rendered.

III. IMPLEMENTATION

The proposed system is a blend of different technologies such as django, A-Frame and MySQL. Django, on one end is the web development framework that was used for deploying VISER. On the other hand, A-Frame is a web framework used for creating virtual reality experiences. MySQL was used for database purposes.

The entire system can be broken down among three different layers, namely the model, template and view. This is the fairly popular Model Template View (MTV) architecture that django follows. Furthermore, each of the earlier mentioned technologies; A-Frame and MySQL can be respectively mapped to the template and the model. As django uses python for development, NLTK was readily available for text processing. As observed from Fig. 1, the flow of data among django takes place in the following manner: firstly, the user enters a textual input in the WebVR world. This virtual world is nothing but the template which is the front end for the user to use. After this, the data previously entered is passed to the view viz. the backend where all the text processing stuff happens to take place such as processing of the context, extracting the spatial relations and getting the other worldly information for the same. Following this processing, the objects that are needed to be rendered in case of DOG or the actions that are to be performed in case of humanoid actions are required to be retrieved from the model. This model forms the last piece of the MTV trio which in simple terms is just the database used. The database contains different tables for storing 3D objects and for storing the verbal animations. The data from these object repositories are passed to the view that were requested in the first place. The view then finishes the processing and finally forms a template or in this case a scene in VR world wherein the results are displayed. The view acts as a controller that calls respective computation for the module in use and then passes the results, literally 'controlling' the entire flow.

Focusing on the template, the VR is created using A-Frame as mentioned earlier. A-Frame is completely open source and free to use. It is written using HTML and provides an entity-component relationship that derives its power from three.js library. A-Frame was specifically chosen for WebVR in the proposed system as it was light weight and powerful. At the same time, it was targeted for WebVR and therefore, it did not touch the browser engine. The response time for something to animate or render was comparatively low and gave a smooth user experience while navigating in the VR world. As compared to that of VRML/ X3D, A-Frame provided a richer, cleaner and optimized reason for the becoming the WebVR choice. Additionally, control features for different headsets are directly available in A-Frame.

The MySQL database was the choice of preference as the schema structure was fixed for storing information of an object. This information is nothing but the data about the 3D objects which includes their position, orientation, animations, scale and size category along with their file names for the files that are stored in the django server that can be fetched as and when required.

3D objects and animations were rendered in the world. The 3D objects are collected from Google Poly which is an open source website for downloading and using readymade objects. But when these objects were used, their origin was mismatching with the origin of the world that we were creating. Other problems also arose such as, the apple was bigger in size than a truck, the table's rotation was absurd which made it look upside down. To solve this issue, each object was tested with reference to some datum which in our case was three objects of different sizes, an apple, a table and an elephant. After normalizing the objects, they were mapped with their real-world counterparts such that a laptop was bigger in size than an apple and a car was smaller than an elephant. These 3D objects are a combination of object (.obj) file and material (.mlt) file along with their textures (.png/ .jpg) if present. Another major problem with these objects was that some were being rendered without any textures or appearing to be black. The material file of these objects was manipulated for not only addressing the problem but also creating new objects out of the original. In such a way, a red apple was manipulated to create apple of twelve other colors.

As for the animations for the humanoid actions, were once again readily available to use from Mixamo open source website. Here the avatars and their actions were downloaded and used in the form of filmbox (.fbx) format as GL Transmission Format (.gltf) and Collaborative Design Activity (.collada) formats were proving to be futile because only collada format was available to use other than the fbx format, A-Frame required conversion of collada models to gltf as the animation mixer used by A-Frame does not support collada format. But the converter was converting the gltf format only for a few objects correctly, fbx on the other hand needed no conversion and was ready to use even though gltf models are just a slightly faster to load.

IV. PERFORMANCE EVALUATION USING SYNTAX RULES

As we cannot directly evaluate a VR-Domain project, So we have conducted our performance evaluation mostly based on the output that is being generated by the NLP Engine. In other words how correctly the 3D scene is being rendered according to the different types of input.

Note: Following are the abbreviations used in formulating the syntax rules:

- NNP - Proper Noun
- VBG - Verb
- NN - Noun
- JJ - Adjective
- PRP - Pronoun

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Syntax rules present different types of English sentences, which comprises of different number of nouns and adjectives associated with it, which when processed by NLP engine renders the correct output.

Following are the syntax rules which have been defined for the module humanoid actions. The input query in the example column is equivalent to the syntax rule column in the table.

TABLE I: SYNTAX RULE FOR HUMANOID ACTION MODULE

Sr. No.	Syntax Rule	Example
1.	NNP VBG	Mohan is dancing
2.	(NNP VBG) + (NNP VBG)	Mohan is dancing, and Maya is running
3.	(NNP VBG) + (VBG PRP)	Mohan is dancing and running is being performed by me
4.	(NNP VBG) + (VBG NNP)	Maya is running, and dancing is being performed by Mohan
5.	(NNP VBG) + (NNP VBG) + (PRP VBG)	Akshay is kicking, and Mohan is dancing. On the other hand, I am punching
6.	(PRP VBG) + (NNP VBG) + (NNP VBG) + (VBG NNP)	I am running, and Mohan is dancing. On the other hand, Akshay is punching and rolling is being performed by dadi

Following are the syntax rules for Dynamic Object Generation module.

TABLE II: SYNTAX RULE FOR DYNAMIC OBJECT GENERATION MODULE

Sr. No.	Syntax Rule	Example
1.	NN JJ JJ	The apple is red in color, and it is big
2.	NN JJ	The apple is on the right/left side
3.	NN	Show me a zebra/computer/spoon etc.
4.	JJ JJ NN JJ	Show me a big red apple which is on the left side
5.	JJ NN JJ	Show me a big apple which is close/closer to me
6.	(NN) + (NN)	Show me an apple and an elephant
7.	(NN) + (JJ NN)	Show me an apple and a big brown table
8.	(JJ NN JJ) + (JJ JJ NN JJ)	Show me a red apple which is on the left side of the big brown table
		which is away
9.	(NN JJ) + (NN JJ)	Show me an apple on the left side and an elephant which is on the right side
10.	(JJ JJ NN JJ JJ) + (JJ JJ NN JJ JJ)	Show me a big brown apple on the left side away from me, and a small brown table which is above and away from me
11.	(NN) + (NN) + (NN JJ)	Apple kept on the table, and the car is on the left side
12.	(NN) + (NN) + (NN)	Show me apple, car and a table

V. RESULTS

The results of the various modules and their representation in the open world are shown here.

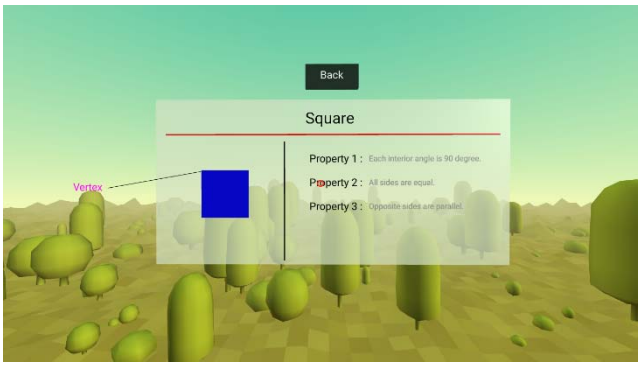


Fig. 3. Courses - Shapes being taught in a 3D world

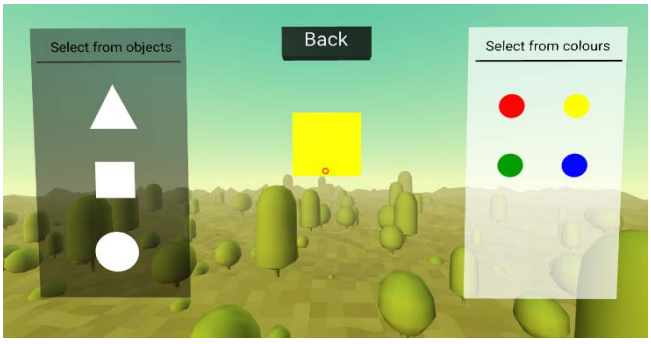


Fig. 4. Courses – Colors being taught in 3D world

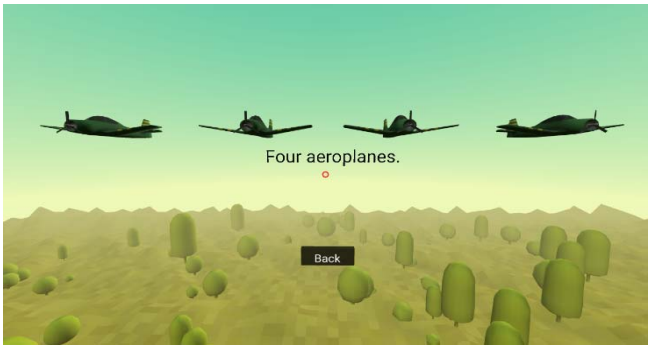


Fig. 5. Courses – Quantity being taught in 3D world

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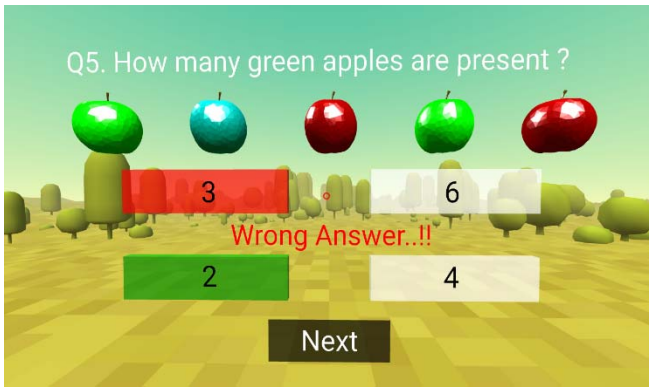


Fig. 6. Quiz being taken on the taught courses by making use of daily objects



Fig. 7. Way of entering textual input in the VR world

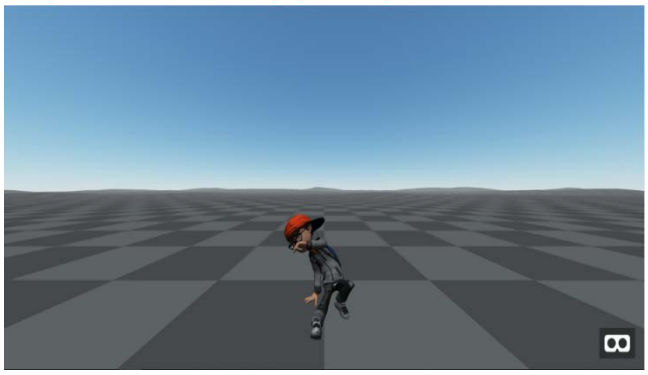


Fig. 8. Humanoid actions- Output of the entered query 'Mohan is dancing'

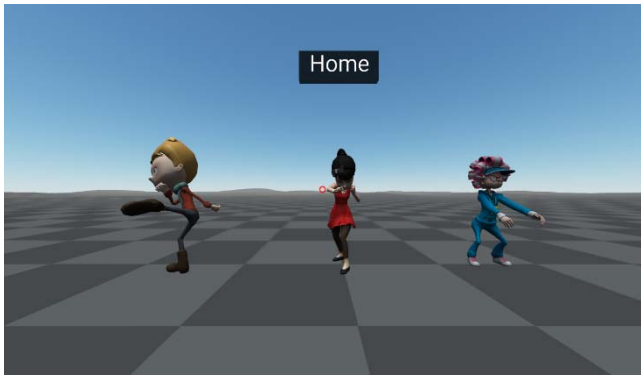


Fig. 9. Humanoid actions- Output of the entered query 'Akshay is kicking. On the other hand, Maya is performing the punching action and Dadi is doing some exercise.'

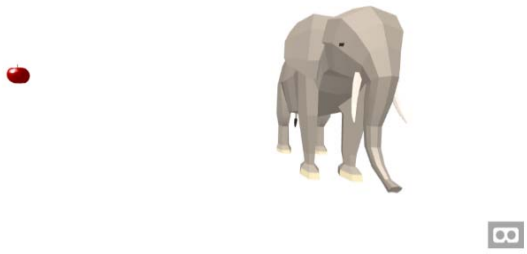


Fig. 10. DOG- Output for the query 'Show me a big red apple on the left side and an elephant on the right side and away from me.'

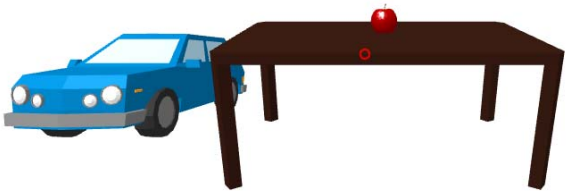


Fig. 11. DOG- Output for the query 'An apple is kept on top of the table. On the other hand, a car is present on the left side which is away from me.'

Note that for Fig. 10, the elephant does not seem to be far away from the user because A-Frame uses a fish eye view for presenting the VR world. Due to this the screen seems to warp around the edges but when we roam around or look at it from the sides, it is actually away from the user.

VI. CONCLUSION AND FUTURE WORK

This paper has given a detailed and in-depth description of the proposed system VISER and how it can used to tackle the issue of dyslexia hindering the education of students suffering with the same. The paper also focuses on how textual inputs are processed to render 3D scenes or animations by formation of tuple method and how education for such students can be carried out by making them learn in simulated 3D world that maps to the real one.

For future work, as the UI is a 3D world type in nature, the extensibility is very high and endless courses can be created by referring the actual curriculum for educating dyslexic students and mapping it in VISER. Currently, the system is using the traditional desktop style WebVR, but the system can be altered for being used along with Head Mounted Devices (HMD) such as Google cardboard, HTC Vive, etc. The textual input can also be replaced by using speech input.

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